RK UNIVERSITY
SCHOOL OF ENGINEERING

B. TECH - SEMESTER–IV

Assignment-2

Subject Code: **APS412** Branch: **CE/IT**

Subject Name: **Discrete mathematics and numerical method** Semester: **4th**

Date of Assignment: **13/02/26** Last date of submission: **10/03/26**

UNIT-1

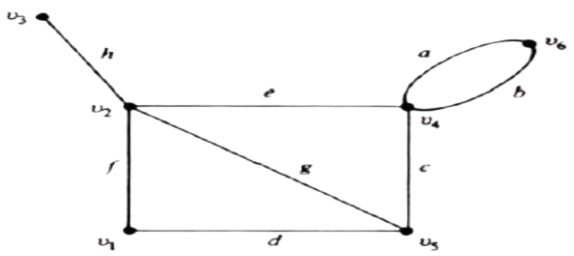
1 A graph G has 15 edges, 3 vertices of degree 4 and other vertices of degree 3. Find the number of vertices in G.

2 Define Complete graph and Regular graph.

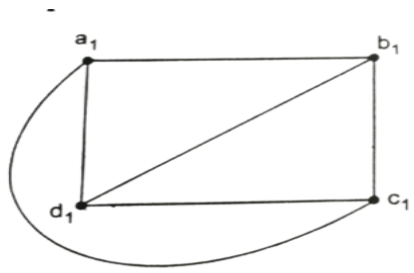
3 Find the number of edges in a r-regular graph with n vertices.

4 Define cycle with examples.

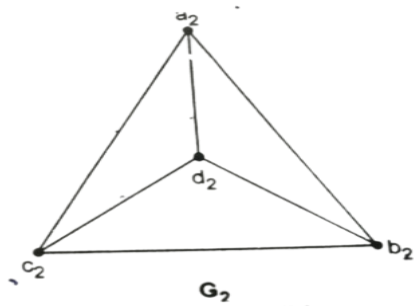
5 Find incidence matrix for the following graph



6 Check whether the following graphs are isomorphic or not.

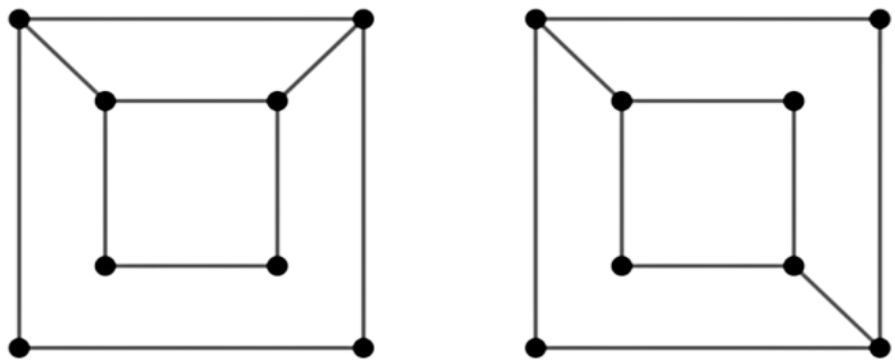


G_1



G_2

7 Define Isomorphic Graphs. Verify the following graph are isomorphic or not.



8 Define tree. Draw all possible tree for six vertices.

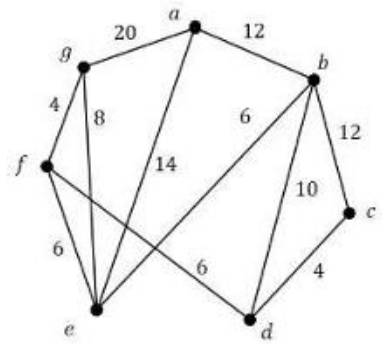
9 Draw graph of the following matrices.

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 1 & 2 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 & 2 & 3 & 2 \\ 1 & 0 & 3 & 5 & 1 \\ 2 & 3 & 2 & 2 & 3 \\ 3 & 5 & 2 & 0 & 4 \\ 2 & 1 & 3 & 4 & 6 \end{bmatrix}$$

$$c = \begin{bmatrix} 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

10 Find the incidence matrix and adjacency matrix for following graph.



11 What is the smallest integer n such that the complete K_n has at least 500 edges?

12 Define following with examples.

1. Simple graph
2. Null graph
3. Regular graph

UNIT-5

Solve following Examples:

1 Find a root of the following equations, correct up to three decimal places by **Bisection Method.**

(i) $x^3 - 4x - 9 = 0$ Ans.= 2.706

(ii) $x^4 - x - 10 = 0$ **(Four Decimal)** Ans.=1.8125

(iii) $cosx = xe^x$ Ans.= 0.5312

(iv) $x - cosx = 0$ **(Two Decimal)** Ans.=0.75

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2 Find a root of the following equations correct up to three decimal places using the **False position method (Regula Falsi method)**:

(i) $x^2 - \log_e x - 12 = 0$ (ii) $x^3 - 2x - 5 = 0$ Ans.= 2.094

(iii) $cosx - 3x + 5 =$ Ans.= 1.642

3 Use **Secant method** to find the roots of $cosx = xe^x$ correct up to three decimal places between 0 and 1. Ans.=0.518

4 Find the root of the equation $x^3 - 2x - 5 = 0$ using **Secant method** correct up to four decimal places. (Using $x_0 = 2, x_1 = 1$) Ans.=2.0946

5 Derive **Secant method** and solve $xe^x - 1 = 0$ correct up to three decimal places between 0 and 1.

6 Find a root of the following equations, correct to three decimal places using **Newton- Raphson method.**

(i) $\text{xsinx} + \text{cosx} = 0$. Ans.= 2.798 (ii) $x \log_{10} x - 1.2 = 0, x > 0$.**(Four Decimal)** Ans.= 2.7406

(iii) $x^4 - x^3 + 10x + 7 = 0$, $[-2, -1]$ $Ans. = -1.454$

(iv) $x^3 - 3x + 5 = 0$ $Ans. = -2.279$

7 Find the $\sqrt[3]{58}$ correct up to three decimal places by using N-R Method. Ans.=3.8708

8 Find the $\sqrt[2]{10}$ correct up to three decimal places by using N-R Method. Ans.=3.162